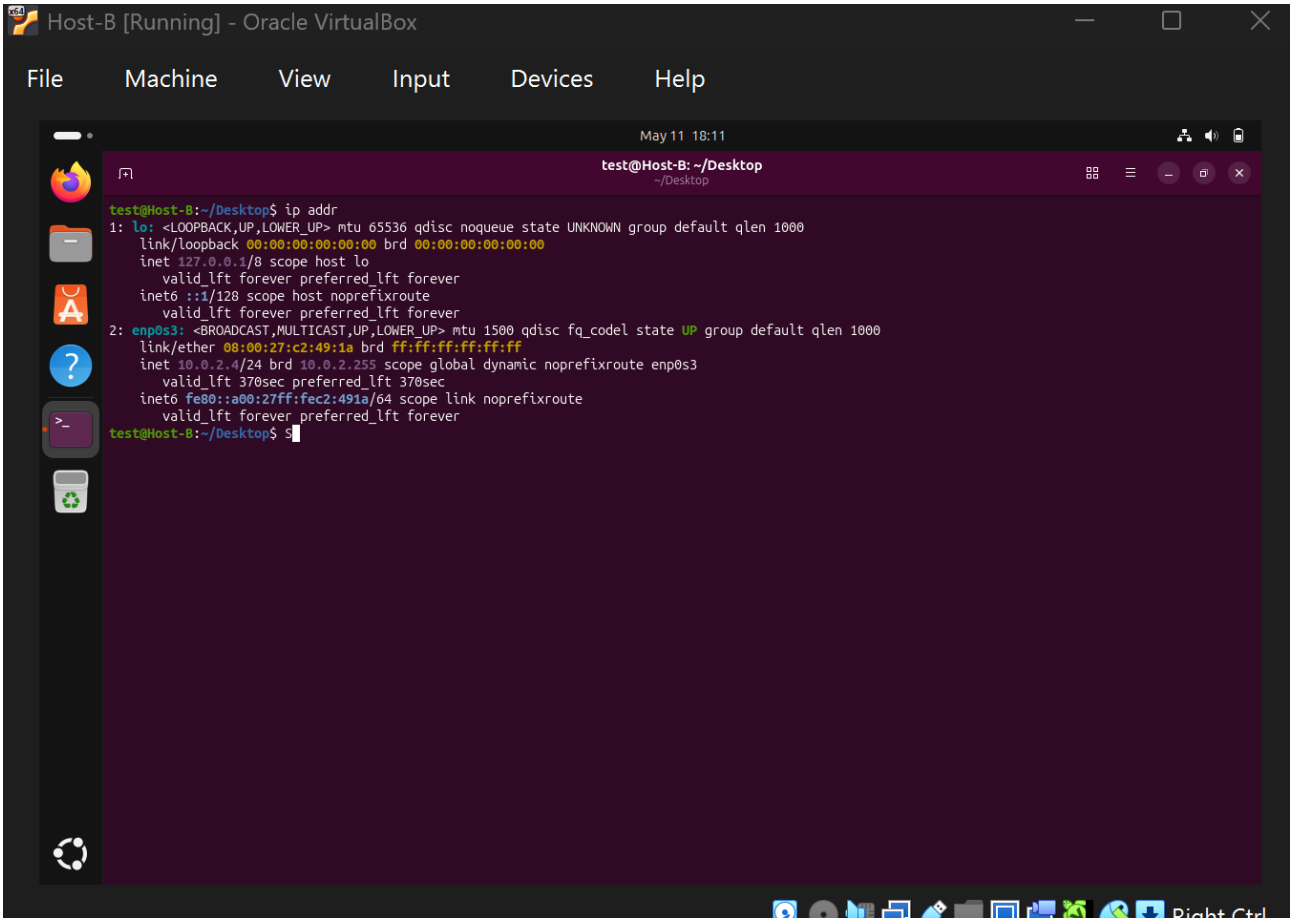
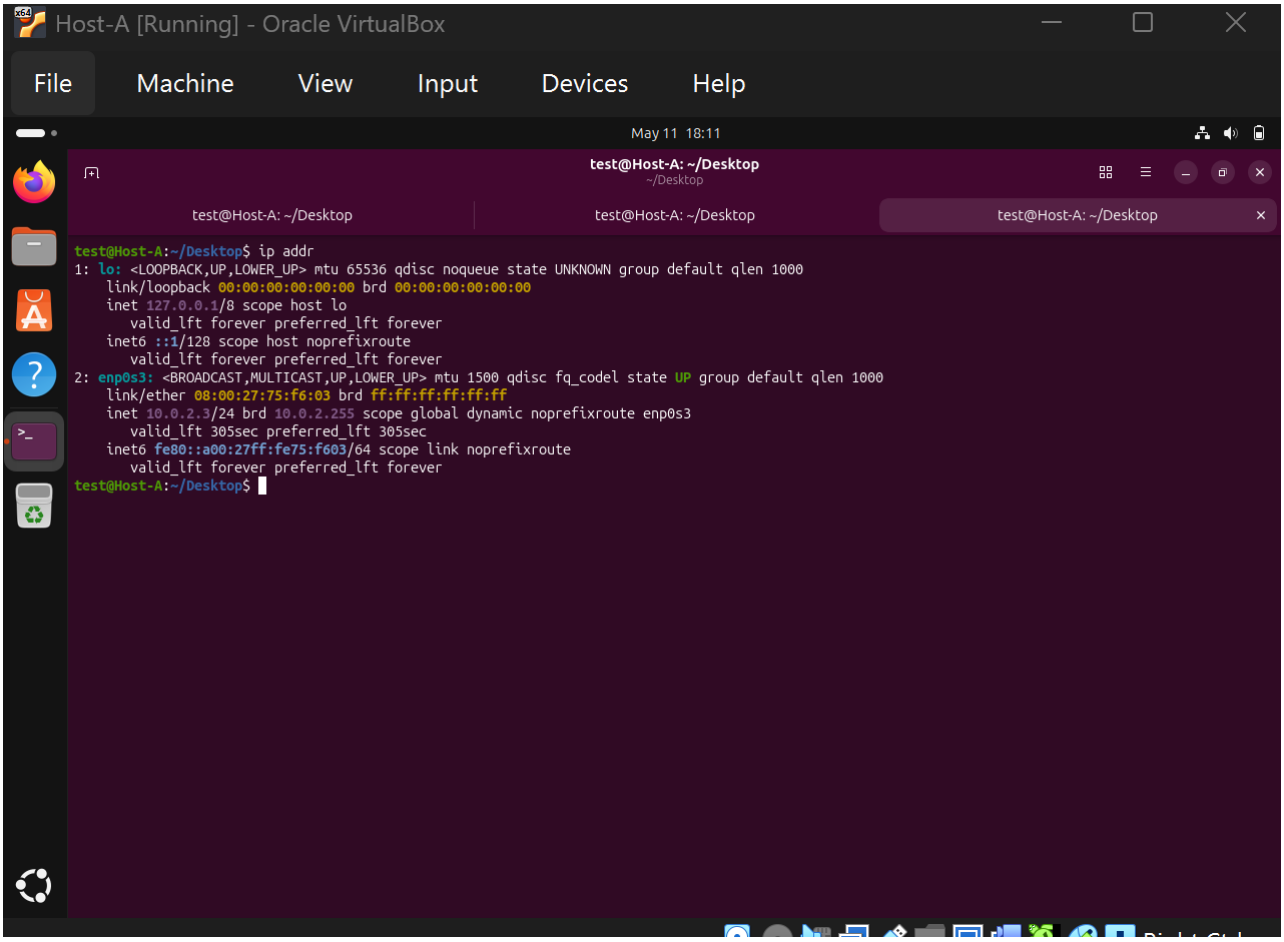
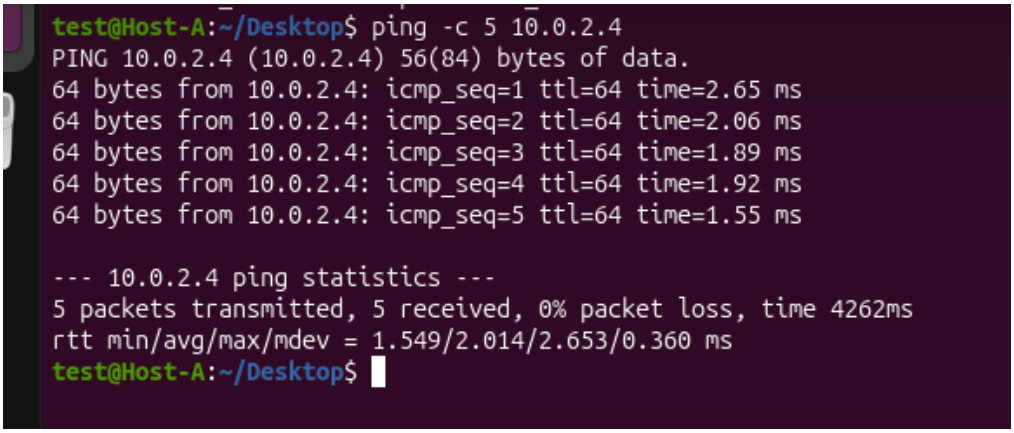
Linux sysctl Network Hardening Lab

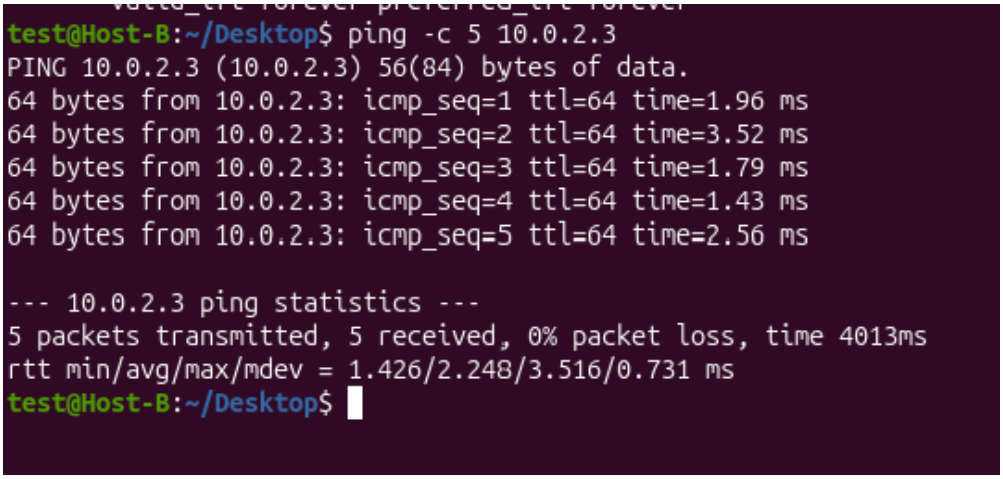
Two VMs configured:





Successful pinging:



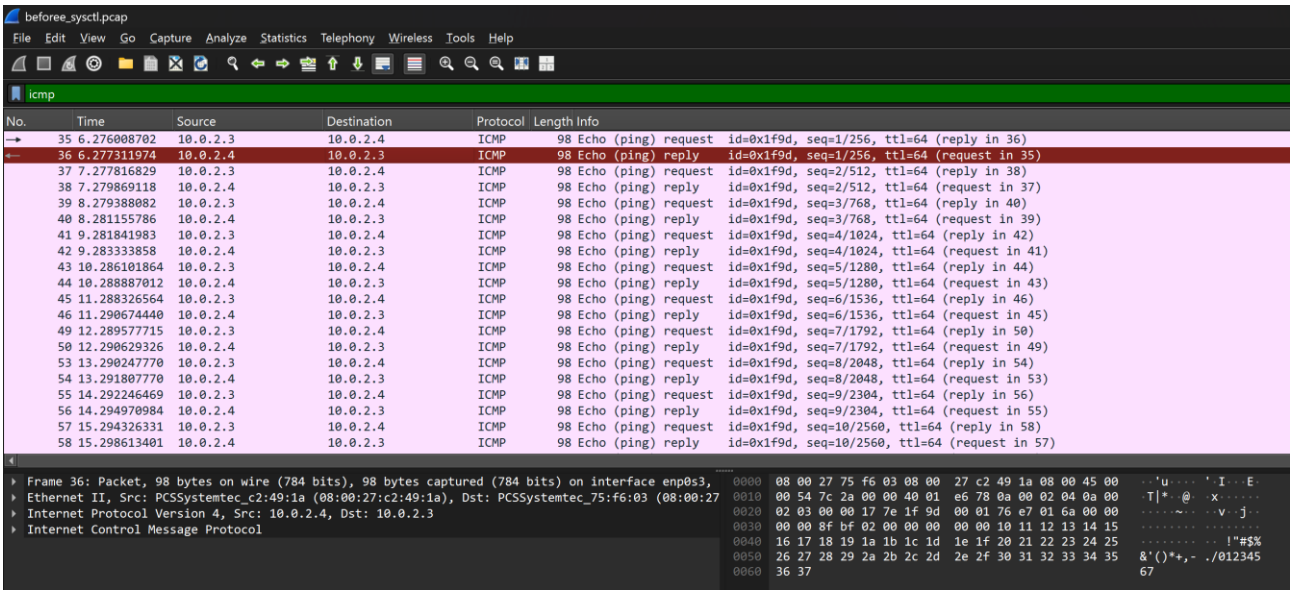


Apply sysctl Hardening (Host B):

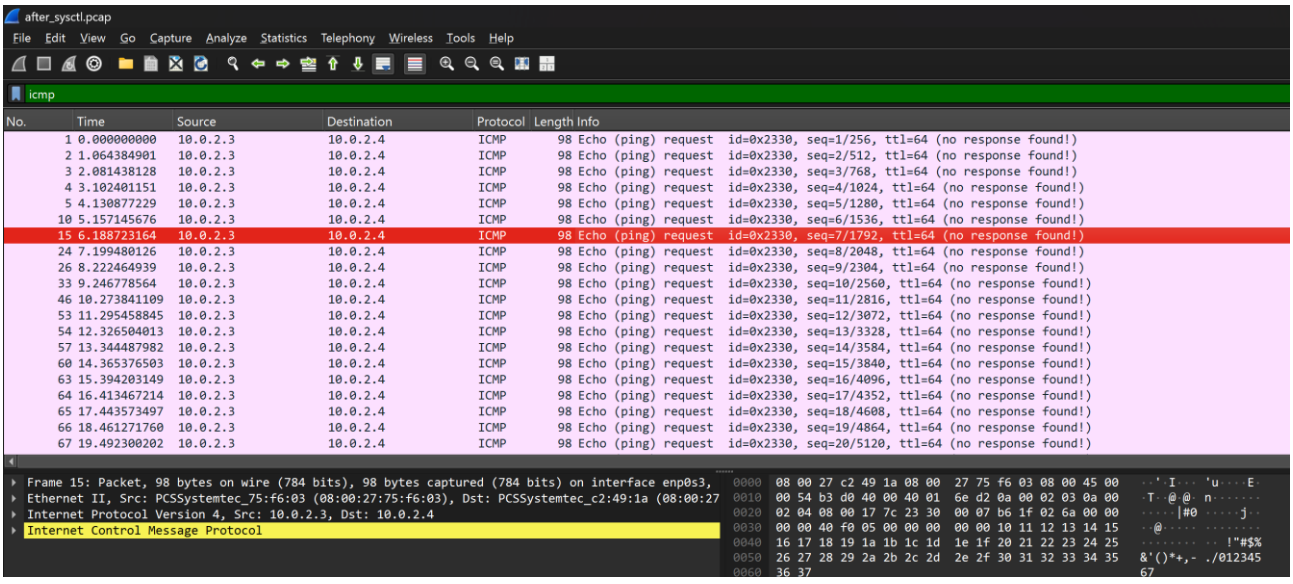
```
--- 10.0.2.3 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4013ms
rtt min/avg/max/mdev = 1.426/2.248/3.516/0.731 ms
test@Host-B:~/Desktop$ sudo nano /etc/sysctl.conf
[sudo: authenticate] Password:
test@Host-B:~/Desktop$ sudo sysctl -p
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.icmp_echo_ignore_all = 1
test@Host-B:~/Desktop$
```

Analysis (Wireshark):

Before:



After:



What is sysctl?

sysctl is a tool in Linux used to view and modify **kernel parameters** at runtime.

The "kernel" is the core of the operating system that manages hardware and networking. Instead of needing to rewrite code or restart the computer, sysctl allows administrators to change how the networking stack behaves on the fly.

What is a Smurf Attack?

A **Smurf Attack** is a type of Distributed Denial of Service (DDoS) attack that uses a technique called **amplification**.

- **Spoofing:** The attacker sends an ICMP Echo Request (ping) to a network's **broadcast address** (`10.0.2.255`).
- **The Trick:** The attacker "spoofs" the source IP address, making it look like the request came from the victim's machine instead of the attacker's.
- **Amplification:** Every single device on that network receives the broadcast ping and sends an Echo Reply back to the victim.
- **Result:** The victim's computer is flooded with thousands of unwanted replies at once, crashing its network connection.

Why are ICMP replies missing after hardening?

- **net.ipv4.icmp_echo_ignore_all = 1**: This is the "Stealth Mode" switch. It tells the Linux kernel to silently drop any ICMP Echo Request packets it receives. The kernel sees the packet, recognizes it's a ping, and simply throws it in the trash without telling the sender.

- **Packet Filtering vs. Service Shutdown**: network card is still physically receiving the data, but the **Operating System** is refusing to process or answer it.